Answers: Introduction to differentiation and the derivative

Sara Delgado Garcia

Summary

Answers to questions relating to the guide on introduction to differentiation and the derivative.

These are the answers to Questions: Introduction to differentiation and the derivative.

Please attempt the questions before reading these answers!

1.1. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(x^3+5x-3\right)=3x^2+5.$

1.2. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(5x\right)=5.$

1.3. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(-5\sqrt{x}\right)=-5\cdot\dfrac{1}{2}x^{-1/2}=-\dfrac{5}{2\sqrt{x}}.$

1.4. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(-\sin(x)\right)=-\cos(x).$

1.5. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\cos x+5\right)=-\sin(x).$

1.6. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(2\sqrt{x}\right)=2\cdot\dfrac{1}{2}x^{-1/2}=\dfrac{1}{\sqrt{x}}.$

1.7. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(2\ln(2x)+x^5\right)=\dfrac{2}{x}+5x^4.$

1.8. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\ln(5x)\right)=\dfrac{1}{5x}\cdot 5=\dfrac{1}{x}.$

1.9. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(e^{-x}\right)=e^{-x}\cdot(-1)=-e^{-x}.$

1.10. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(23x+5\right)=23.$

1.11. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(4x+100\right)=4.$

1.12. For $\sinh(5x)=\dfrac{e^{5x}-e^{-5x}}{2}$, it follows that

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\sinh(5x)\right)=5\frac{e^{5x}+e^{-5x}}{2}=5\cosh(5x)$$

since $\cosh(x)=\dfrac{e^{x}+e^{-x}}{2}$.

1.13. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\cos(3x)-\sin(2x)\right) = -3\sin(3x) - 2\cos(2x).$

1.14. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\ln(x)+\cos(x)+3x\right) = \dfrac{1}{x} - \sin(x) + 3.$

1.15. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\dfrac{2}{5}\sinh(x) + \dfrac{2}{13}\cosh(x)\right) = \dfrac{2}{5}\cosh(x) + \dfrac{2}{13}\sinh(x).$

1.16. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(e^{5x}+x^2+3\right) = 5e^{5x} + 2x.$

1.17. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\ln(x)+x^2\right) = \dfrac{1}{x} + 2x.$

1.18. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\ln(5x)-\ln(x)\right) = \dfrac{1}{x} - \dfrac{1}{x} = 0.$

1.19. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\cosh(x)-5x^7\right) = \sinh(x) - 35x^6.$

1.20. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{3x^2}\right) = \sqrt{3}$

1.21. $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(x^3+3x-\sqrt{2x}\right) = 3x^2 + 3 - \dfrac{1}{\sqrt{2x}}.$

Version history and licensing

v1.0: initial version created 03/25 by Sara Delgado Garcia as part of a University of St Andrews VIP project.

This work is licensed under CC BY-NC-SA 4.0.